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Program Developed By:

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In [8]:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image = cv2.imread('shravin.jpeg') # Replace with your image path
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Original Image
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')
```

(np.float64(-0.5), np.float64(1599.5), np.float64(1599.5), np.float64(-0.5))

Original Image

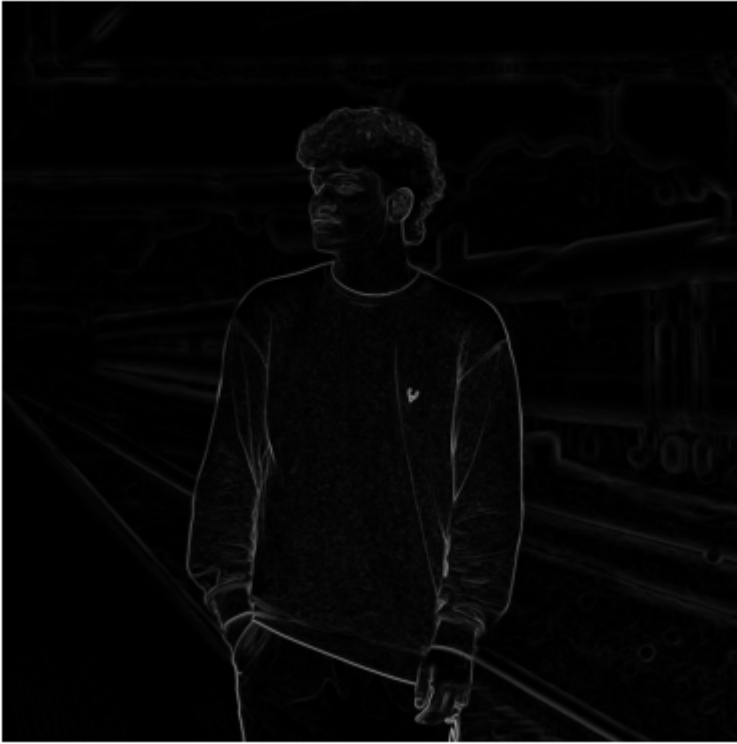


In [9]:

```
sobel_x = cv2.Sobel(gray_image, cv2.CV_64F, 1, 0, ksize=5) # Sobel in x direction
sobel_y = cv2.Sobel(gray_image, cv2.CV_64F, 0, 1, ksize=5) # Sobel in y direction
sobel_combined = cv2.magnitude(sobel_x, sobel_y) # Combine both directions
plt.imshow(sobel_combined, cmap='gray')
plt.title('Sobel Edge Detection')
plt.axis('off')
```

(np.float64(-0.5), np.float64(1599.5), np.float64(1599.5), np.float64(-0.5))

Sobel Edge Detection



In [10]:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Define Prewitt kernels
kernel_prewitt_x = np.array([[ -1,  0,  1],
                             [ -1,  0,  1],
                             [ -1,  0,  1]], dtype=np.float32)

kernel_prewitt_y = np.array([[ -1, -1, -1],
                             [  0,  0,  0],
                             [  1,  1,  1]], dtype=np.float32)

# Apply filters using cv2.filter2D
prewitt_x = cv2.filter2D(gray_image, cv2.CV_64F, kernel_prewitt_x)
prewitt_y = cv2.filter2D(gray_image, cv2.CV_64F, kernel_prewitt_y)

# Combine both directions
prewitt_combined = cv2.magnitude(prewitt_x, prewitt_y)

plt.imshow(prewitt_combined, cmap='gray')
plt.title('Prewitt Edge Detection')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1599.5), np.float64(1599.5), np.float64(-0.5))
```

Prewitt Edge Detection



In [11]:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Define Roberts Cross kernels
kernel_roberts_x = np.array([[ 1,  0],
                             [ 0, -1]], dtype=np.float32)

kernel_roberts_y = np.array([[ 0,  1],
                             [-1,  0]], dtype=np.float32)

# Apply filters using cv2.filter2D
roberts_x = cv2.filter2D(gray_image, cv2.CV_64F, kernel_roberts_x)
roberts_y = cv2.filter2D(gray_image, cv2.CV_64F, kernel_roberts_y)

# Combine both directions
roberts_combined = cv2.magnitude(roberts_x, roberts_y)

plt.imshow(roberts_combined, cmap='gray')
plt.title('Roberts Edge Detection')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1599.5), np.float64(1599.5), np.float64(-0.5))
```

Roberts Edge Detection



In [12]:

```
laplacian = cv2.Laplacian(gray_image, cv2.CV_64F)
plt.imshow(laplacian, cmap='gray')
plt.title('Laplacian Edge Detection')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1599.5), np.float64(1599.5), np.float64(-0.5))
```

Laplacian Edge Detection



In [13]:

```
canny_edges = cv2.Canny(gray_image, 50, 150)
plt.imshow(canny_edges, cmap='gray')
plt.title('Canny Edge Detection')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1599.5), np.float64(1599.5), np.float64(-0.5))
```

Canny Edge Detection

